ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ROBOTICS FUNDAMENTALS

PRACTICE EXAM 3 · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Robotics Fundamentals practice exam 3.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and examples.

SECTION A

QUESTION 1

1.1 1.1.1 A ✓ 1.1.2 B ✓ 1.1.3 C ✓ 1.1.4 D ✓ 1.1.5 B ✓ 1.1.6 C ✓ 1.1.7 A ✓ 1.1.8 A ✓ 1.1.9 B ✓ 1.1.10 D ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 False ✓ 1.2.4 True ✓ 1.2.5 True ✓ 1.2.6 False ✓ 1.2.7 True ✓ 1.2.8 False ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 G ✓ 1.3.3 I ✓ 1.3.4 E ✓ 1.3.5 B ✓ 1.3.6 J ✓ 1.3.7 A ✓ 1.3.8 H ✓ 1.3.9 D ✓ 1.3.10 F ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 A programmable machine that can sense its surroundings, make decisions and carry out tasks (act) automatically ✓✓ (2)

2.2 Any THREE ✓✓✓: Karel Čapek's play R.U.R. (1920) used the word "robot"; Isaac Asimov's three laws of robotics; George Devol and Joseph Engelberger built Unimate, the first industrial robot (1961), used by General Motors; the Mars rovers; the first robot vacuum cleaners; humanoid robots such as ASIMO. (3)

2.3 Any FOUR ✓✓✓✓: wheeled – fast on flat floors; tracked – tank tracks for rough or soft ground; legged / walking – can step over obstacles and climb stairs; flying – drones; underwater / swimming – submarine robots; rolling / spherical robots. (4)

2.4 Any FOUR ✓✓✓✓: work 24 hours without breaks; higher and more even quality; faster production; safer for workers (dangerous jobs); lower long-term costs; very accurate and repeatable; less waste. (4)

2.5 Dull – e.g. repeating the same assembly or packing task ✓; dirty – e.g. cleaning drains, painting, mining ✓; dangerous – e.g. bomb disposal, handling chemicals, exploring volcanoes or space ✓ (3)

[16]

QUESTION 3

3.1 Sense: bump, cliff and dirt sensors detect walls, stairs and dirt ✓; think: the controller decides where to go next ✓; act: the motors drive the wheels and brushes ✓. (3)

3.2 How it works ✓ and use ✓ for each: (a) touch switch – closes a contact when pressed – bumper to detect collisions; (b) IR reflective – sends infrared light and measures how much reflects back (black reflects less) – line following, edge detection; (c) ultrasonic – sends a sound pulse and times the echo – measuring distance, avoiding obstacles; (d) LDR – resistance falls as light increases – light-seeking robots, automatic headlights. (8)

3.3 $\mathrm{d} = (\mathrm{v} \times \mathrm{t}) \div 2$ ✓ $= (340 \times 0{,}004) \div 2$ ✓ $= 0{,}68\ \mathrm{m}$ ✓ (3)

3.4 Any TWO differences ✓✓✓✓: a DC motor spins continuously, a servo turns to a set angle and holds it; a DC motor needs only a supply (speed by PWM), a servo needs a control signal; a servo has built-in gears and feedback (potentiometer); DC motors suit wheels, servos suit arms, grippers and steering. (4)

3.5 A small gear driving a large gear lowers the speed ✓ and increases the turning force (torque) by the same ratio ✓. (2)

[20]

QUESTION 4

4.1 One independent direction in which a robot or joint can move (move along or turn about an axis) ✓✓ (2)

4.2 Cartesian / gantry ✓, cylindrical ✓, spherical / polar ✓, articulated (and SCARA accepted) ✓. Description of ONE, e.g. Cartesian moves in three straight lines along X, Y and Z ✓ (5)

4.3.1 $\mathrm{v} = \pi \mathrm{dN} \div 60$ ✓ $= \pi \times 0{,}06 \times 100 \div 60$ ✓ $= 0{,}31\ \mathrm{m/s}$ ✓ (3)

4.3.2 $\mathrm{s} = \mathrm{v} \times \mathrm{t} = 0{,}314 \times 10 = 3{,}14\ \mathrm{m}$ ✓ (1)

4.4 Each side has its own motor ✓. To turn left, the right wheel turns faster than the left (or the left wheel stops / reverses) ✓; to turn right, the left wheel turns faster ✓. (3)

4.5 Tracks spread the weight over a large area so the robot does not sink ✓ and they grip loose sand better than narrow wheels ✓. (2)

[16]

QUESTION 5

5.1 Any FOUR ✓✓✓✓: wear safety goggles; switch off and remove the battery before working on the robot; keep fingers away from wheels and gears; test on the floor or a stand, not at the edge of a table; check wiring for short circuits before switching on; keep the work area tidy; use tools correctly. (4)

5.2 Any TWO ✓✓: use the correct charger; never short-circuit the terminals; do not charge unattended or overnight; stop using a swollen, damaged or hot battery; store in a cool place / fire-safe bag. (2)

5.3 Any TWO explained ✓✓✓✓: job losses as robots replace workers; who is responsible if a robot causes harm; privacy – robots with cameras collect personal data; safety of robots working near people; use of robots in weapons; people relying too much on machines. (4)

5.4 Positive: new jobs in programming, maintenance and design / safer work ✓. Negative: some low-skilled jobs disappear ✓. (2)

5.5 Any FOUR ✓✓✓✓: charge and inspect the battery; check that screws and nuts are tight; clean the sensors; check wheels / tracks and gears for wear and hair or dirt; check wires and connectors; test the emergency stop and bumpers; update the software. (4)

5.6 A hacker could take control of the robot or steal its data ✓, which could cause damage or injury, so passwords, updates and secure connections are needed ✓. (2)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE